

AME 323: Mini-Project 1

Taylor-Maccoll solver

Spring 2026

The Taylor-Maccoll equations for supersonic flow over a cone are

$$\frac{\gamma - 1}{2} \left[V_{\max}^2 - V_r^2 - \left(\frac{dV_r}{d\theta} \right)^2 \right] \left(2V_r + \frac{dV_r}{d\theta} \cot \theta + \frac{d^2 V_r}{d\theta^2} \right) - \frac{dV_r}{d\theta} \left[V_r \frac{dV_r}{d\theta} + \frac{dV_r}{d\theta} \left(\frac{d^2 V_r}{d\theta^2} \right) \right] = 0, \quad (1a)$$

$$\frac{dV_r}{d\theta} - V_\theta = 0, \quad (1b)$$

where $V_{\max}$ is the maximum theoretical velocity for a given h_0 . In dimensionless form, Equation 1a is

$$\frac{\gamma - 1}{2} \left[1 - V_r'^2 - \left(\frac{dV_r'}{d\theta} \right)^2 \right] \left(2V_r' + \frac{dV_r'}{d\theta} \cot \theta + \frac{d^2 V_r'}{d\theta^2} \right) - \frac{dV_r'}{d\theta} \left[V_r' \frac{dV_r'}{d\theta} + \frac{dV_r'}{d\theta} \left(\frac{d^2 V_r'}{d\theta^2} \right) \right] = 0, \quad (2)$$

where

$$V' = \frac{V}{V_{\max}}. \quad (3)$$

In groups of 2, your task is to do the following:

1. Write a program in MATLAB®, Python, or another language of your choice that solves the Taylor-Maccoll equations for conical shocks and can calculate the shock angle (β) for a given Mach number (M) and cone semivertex angle (θ_c).
 - (a) It would be helpful to first write a script that solves the δ - β - M equation.
 - (b) You need not write your own integration routine (you can use `ode45` or similar built-in functions).
 - (c) There are multiple ways to solve this problem:
 - i. The shooting method as described in class (and AME 302) is the most straightforward way of doing this, though other solutions exist. An internet search will turn up a number of possible solutions.
 - ii. Converging on the proper solution for β can be achieved a number of ways, such as Newton's method, the bisection method, or by clever use of various built-in MATLAB® functions, among others.
 - (d) **You must turn in your code**, and it may not be a copy of something you found on the internet.
2. Create a plot of the shock-wave angle, β , versus the cone angle, θ_c in air similar to the one in Chart 5 of *NACA 1135* (pp. 48–49).



- (a) Include curves for $M_\infty = 1.5, 2.0$, and 5.0 .
 - (b) Determine the maximum cone angles for each of the above values of M_∞ before a detached shock forms. How do these compare with their counterparts for flow over a wedge?
3. Prepare a short user manual for your solver (maximum 6 pages) that includes the following:
- (a) a description of what your code does, how it operates, and how to use it (you can include additional figures if it helps and they will not count against the page count, but this is not necessary);
 - (b) a full page figure of β vs. θ_c plot from above (not included in page count) and discussion of how cone flows differ from wedge flows;
 - (c) in-text citations of and a references/bibliography section containing any sources you use to produce your report other than the course notes (not included in page count); and
 - (d) an acknowledgment and discussion of any use of AI/LLMs to solve the problem.

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